AI Apps & Agents Dev Days

Build Secure, Observable, Production-ready Agents with a Control Plane

Microsoft | NVIDIA

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Agenda

1. **Microsoft & NVIDIA Partnership** — Decades of joint innovation
2. **NVIDIA Integrations on Azure** — GPU infrastructure & AI services
3. **NVIDIA Nemotron Models** — Open models for trustworthy agentic AI
4. **Control Plane Architecture** — Foundry Agent Service + GPU agents on ACA
5. **Document Processing Scenario** — Multi-model pipeline with security & observability
6. **Live Demo** — End-to-end agent orchestration
7. **Key Takeaways & Resources**

NVIDIA Platforms on Azure

NVIDIA GB200, GB300 NVL72

NVIDIA DGX Cloud

NVIDIA H100, H200, B40

NVIDIA Omniverse on DGX Cloud

NVIDIA InfiniBand Networking

NVIDIA AI Enterprise

Confidential Computing

NVIDIA NIM

NVIDIA Accelerates Azure

Azure OpenAI Service
NVIDIA H200

Azure AI Foundry
NVIDIA NIM & AI-Q

Microsoft 365 Copilot
NVIDIA Dynamo-Triton, cuVS

Azure Container Apps
NVIDIA A100

Azure Databricks
NVIDIA RAPIDS

Microsoft Teams
NVIDIA Dynamo-Triton

Azure Confidential Computing
NVIDIA H100

Azure Kubernetes Service
NVIDIA AI Enterprise

Windows 365
NVIDIA RTX vWS

Azure Machine Learning
NVIDIA TRT-LLM

Azure Maps
NVIDIA cuOpt

Microsoft Bing
NVIDIA CV-CUDA, cuVS

NVIDIA Nemotron

Open Family of AI Models, Datasets and Techniques — Build Trustworthy AI Agents

Models

Nano, Super, Ultra
1000+ derivatives
Download from HuggingFace
or run as NIM anywhere

Data

9T tokens, 30M samples
1M compute hours
Full transparency with open datasets

Libraries

NeMo-RL, Minitron, NAS
Optimize for application and compute needs

Research

200+ papers
Open recipes to build right-sized models

Open-weight models • Open datasets • Open recipes → Trustworthy, production-ready agents

What Does Nemotron Bring to Agentic AI?

Open Models

NVIDIA's open data and optimization techniques ensure powerful, transparent, and adaptable models for developers and enterprises. Models and training data are published openly on Hugging Face.

High Compute Efficiency

Through the pruning of larger models, the Nemotron family is optimized for top compute efficiency, using NVIDIA TensorRT™-LLM to deliver higher throughput and on-or-off reasoning capabilities.

High Accuracy

Built on popular open reasoning models for their exceptional knowledge, post-trained with high-quality training data, and aligned to reason like humans, Nemotron models achieve the highest accuracy on leading benchmarks.

Secure and Simple Deployment

The Nemotron model family, available as optimized NVIDIA NIM™ microservices, offers peak inference performance and flexible deployment options, ensuring superior security, privacy, and portability.

https://developer.nvidia.com/nemotron

NVIDIA Nemotron Models for Diverse Workloads

Reasoning

Select from a range of Nemotron reasoning models —**Nano** provides superior accuracy for PC and edge devices, **Super** offers the highest accuracy and throughput to run on a single NVIDIA Tensor Core GPU, and **Ultra** delivers the best accuracy for complex systems optimized for multi-GPU data centers.

Visual Understanding

Nemotron models deliver leading accuracy in multi-image reasoning and video understanding, with strong document intelligence, visual Q&A, and summarization capabilities.

Use NVIDIA Nemotron Parse to extract document data, then NVIDIA Nemotron 2 Nano VL to correlate textual content with charts, diagrams, and other visuals for intelligent querying

Speech

NVIDIA Nemotron Speech models provide high-throughput, ultra-low latency automatic speech recognition (ASR), text-to-speech (TTS), and neural machine translation (NMT) for agentic AI applications.

Retrieval-Augmented Generation

Nemotron RAG models deliver fast, accurate document understanding by extracting multimodal structured information, generating high-quality embeddings, and rank-ordering the most relevant documents. They provide scalable, high-speed retrieval that enhances data quality for LLM training, boosts agent and retriever performance, and streamlines document workflows.

Safety

NVIDIA Nemotron Safety models provide real-time protection against harmful content, off-topic drift, and jailbreak attempts. They add a multilingual, multimodal, content safety layer with reasoning capabilities, enhancing moderation and ensuring cultural alignment.

NVIDIA Nemotron Advances Agentic AI

Most Open and Efficient Models for Agentic AI

Leading Accuracy

Excels at complex reasoning, instruction following and diverse agentic tasks

Highest Throughput

Thinks faster for more reasoning and scales to lower inference cost

Open Source

Trained with commercially viable data Open tools, techniques and datasets

Enterprise-Grade

Secure, stable, supported for production deployments

Faster Time-To-Market

Simplify development with NIM, NeMo, Blueprints

The Azure AI Platform with NVIDIA Technologies

	Build Apps & Agents	Data Strategy	Model Selection & Customization	Deployment	Operations & Security
Microsoft	**GitHub Copilot** Build software faster with AI assistance	**Microsoft Fabric** Unified analytics and AI scaling	**Microsoft Foundry** End-to-end AI app development and scaling	**Azure AI Infrastructure** Scalable, optimized compute for AI workloads	**Azure Content Safety** Built-in AI safety and moderation
	Visual Studio Build and manage production-ready apps	**Azure Databases** Scalable, low-latency AI data stores	**Microsoft Foundry Models** Pre-trained models for faster agent builds	**Azure Kubernetes Service (AKS)** Orchestrate and scale AI agents with Kubernetes	**Microsoft Entra ID** Secure identity and access for agents
	Visual Studio Code Quickly develop and customize applications		**Microsoft Foundry Tools** Integrated tools to build and operate agents	**Azure Container Apps** Fully managed cloud-native containers for agents	**Microsoft Defender** Protect AI workloads from threats
	Copilot Studio Design, deploy, manage copilots & agents		**Azure Machine Learning** Managed training and deployment automation	**Azure App Service** Quickly build, deploy, and scale web apps and APIs	**Azure Monitor** Observe, troubleshoot, & optimize AI systems
NVIDIA	**NVIDIA AI Blueprints** Accelerate AI projects using proven blueprints	**NeMo Curator** High-quality training data curation	**NVIDIA Nemotron** Select and customize open foundation models	**NVIDIA NIM** Containers with optimized models & for inferencing	**NeMo Guardrails** Policy-based controls for LLM behavior
	NeMo Agent Toolkit Build, connect, and scale intelligent agents	**NeMo Data Designer** Synthetic data design for AI apps and agents	**NeMo Evaluator** Run various benchmarks & agent evaluation jobs	**NVIDIA NeMo Microservices** Production-ready LLM deployment services	**NeMo Auditor** Audit, evaluate, and validate AI outputs
		NeMo Safe Synthesizer Safe synthetic data generation	**NVIDIA NCG** Enterprise services, software, and support	**NVIDIA AI Enterprise** Enterprise AI software, support, and security	
		NeMo Retriever Low-latency retrieval for RAG		**NVIDIA Run:AI** Accelerated AI workflows with dynamic orchestration	
				NVIDIA Dynamo Distributed AI inference	

The Challenge: From Prototype to Production

What Breaks at Scale

- ✗ Agents call each other without access controls
- ✗ No visibility into multi-step GPU workloads
- ✗ Security policies applied inconsistently
- ✗ No audit trail for agent decisions
- ✗ Debugging failures across model boundaries

What a Control Plane Provides

- ✓ Clear separation of control and execution
- ✓ Centralized policy enforcement
- ✓ End-to-end observability across agents
- ✓ Consistent security at every boundary
- ✓ Auditable, explainable agent decisions

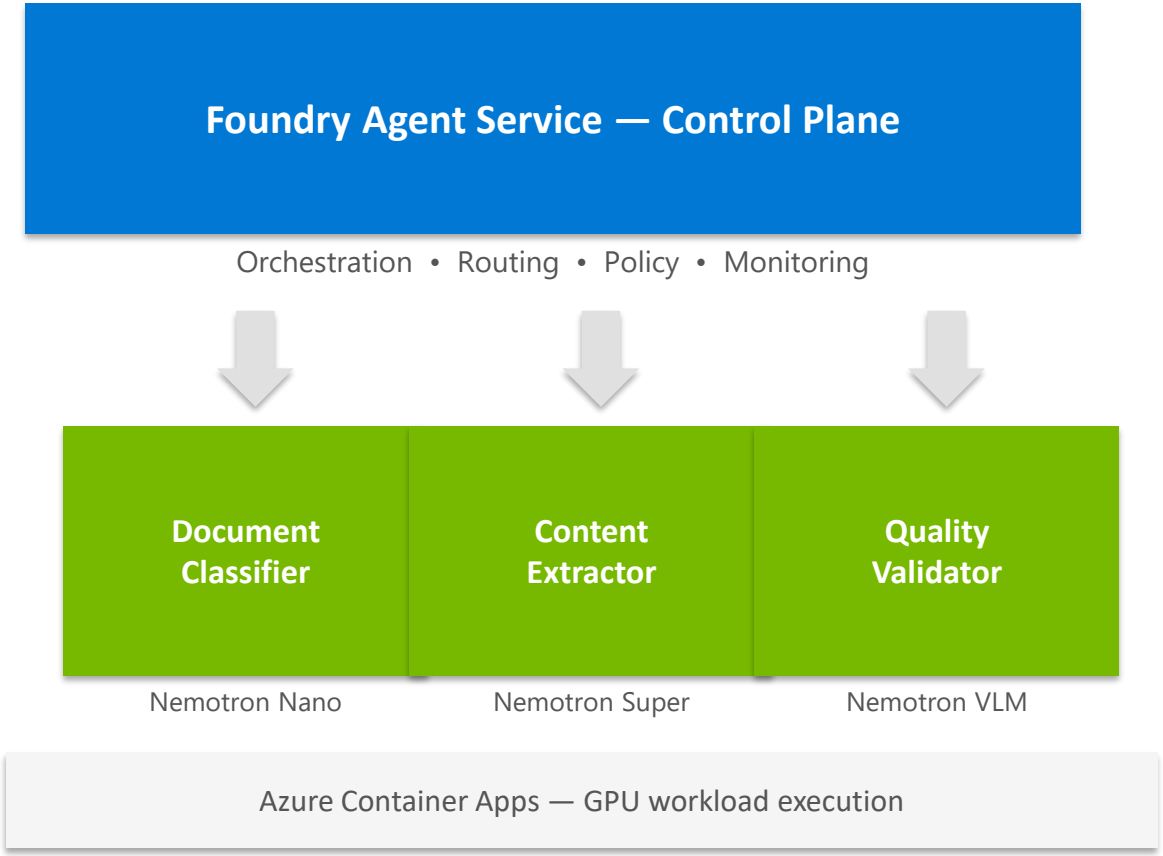
Architecture: Foundry Control Plane + GPU Agents

Security Controls

mTLS between agents
Token-scoped access
Request validation
Audit logging

Observability

Distributed tracing
Token usage metrics
Latency per agent
Error correlation



Data Flow

1. Document ingested
2. Classified by type
3. Content extracted
4. Quality validated
5. Results returned

Scenario: Document Processing with Multiple Models

Using NVIDIA Nemotron models for classification, extraction, and validation

Ingest

Documents arrive via API gateway

Classify

GPT4o verifies doc type

Extract

nemotron-parser verifies and extracts content

Validate

GPT4o quality check

Return

Structured results with audit trail

Every step is: secured • traced • policy-enforced • auditable

Security at Every Hop

Each agent-to-agent call authenticated and authorized

Consistent Tracing

Correlation IDs propagated across all GPU workloads

Explainable Results

Full decision chain from input to output

Security: Zero-Trust Agent Communication

Identity & Access

Managed identity for each agent
Token-scoped permissions
No shared secrets

Network Security

Private endpoints between services
mTLS for agent-to-agent calls
Network isolation per workload

Policy Enforcement

Request/response validation
Content safety filtering
Rate limiting per agent

Audit & Compliance

Complete decision audit trail
Immutable logging
Regulatory compliance ready

Defense-in-depth: every layer enforces security independently

Observability: End-to-End Agent Monitoring

Distributed Tracing Across GPU Agents

- → Correlation IDs span the full agent chain
- → Per-agent latency and token consumption
- → GPU utilization metrics per workload
- → Model inference timing (TTFT, TPS)
- → Error propagation and root cause analysis
- → Request/response payload logging (configurable)

Monitoring Stack

Azure Monitor
Centralized metrics & alerts

Application Insights
APM for agent performance

Log Analytics
KQL queries across agent logs

OpenTelemetry
Vendor-neutral instrumentation

Dashboards
Real-time agent health views

Demo

Document Processing with Foundry Control Plane
and GPU-Accelerated Nemotron Agents

- Agent registration and discovery
- Secure request routing across GPU workloads
- Real-time tracing and observability
- Policy enforcement in action

Key Takeaways

1. **Separate Control from Execution**
   Foundry Agent Service orchestrates while GPU agents execute — clear boundaries enable security and observability

2. **Open Models, Production Quality**
   NVIDIA Nemotron provides open-weight models with enterprise-grade accuracy for agentic workloads

3. **Security by Design**
   Zero-trust communication, identity-scoped access, and policy enforcement at every agent boundary

4. **Observable from Day One**
   Distributed tracing, token metrics, and GPU utilization monitoring across the entire agent chain

5. **Azure-Native Infrastructure**
   Container Apps for GPU workloads, Foundry for orchestration — production-ready with managed services

Resources & Next Steps

→ Microsoft Foundry Agent Service Documentation

→ NVIDIA Nemotron on Hugging Face & build.nvidia.com

→ Azure Container Apps GPU workload guides

→ OpenTelemetry for AI agent instrumentation

→ Azure AI Foundry Model Catalog — Nemotron models

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https://aka.ms/microsoftnvidiademo

Thank You!

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